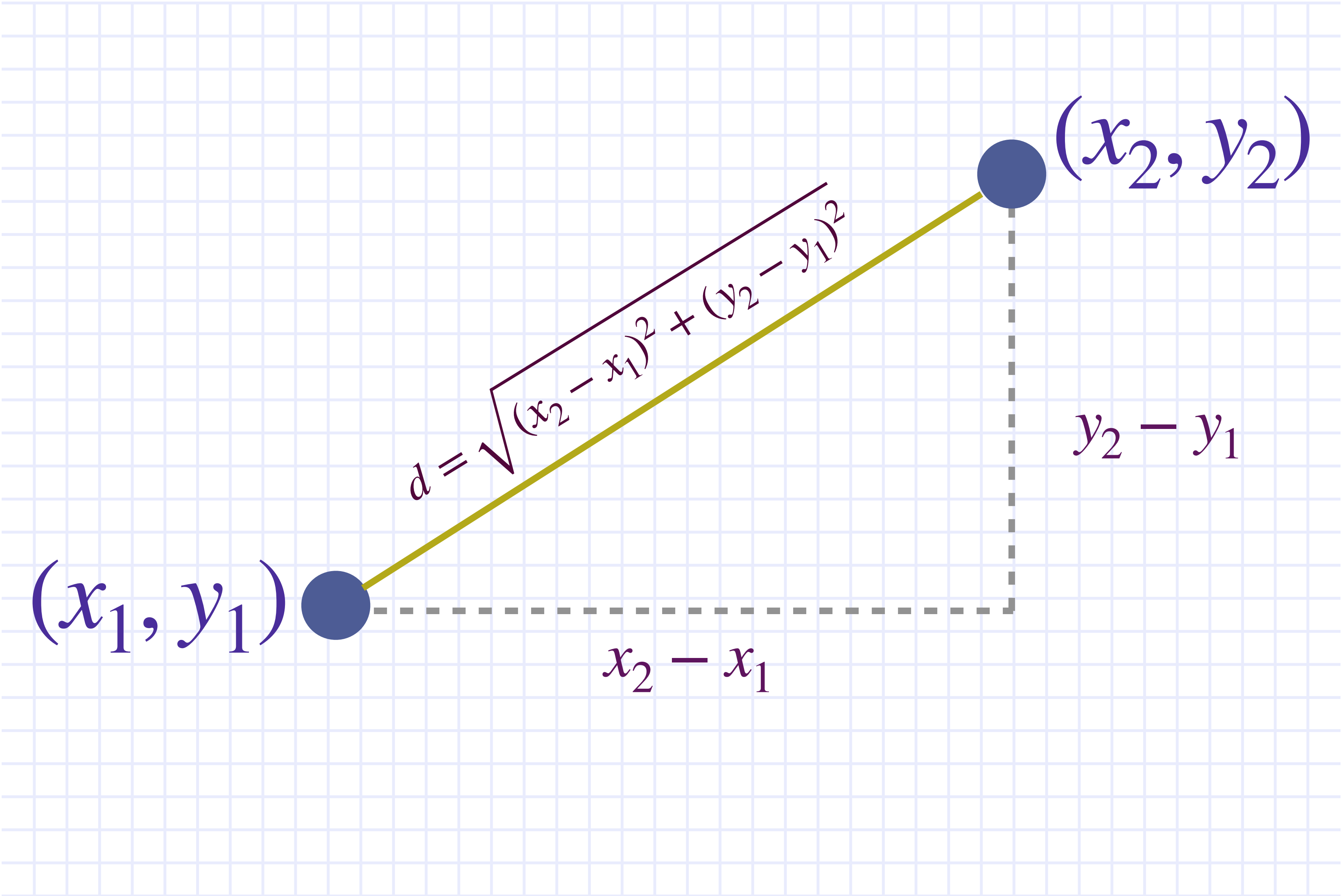
DISTANCE IN KNN (EUCLIDEAN DISTANCE)

For two points in a plane with coordinates (x_1, y_1) and (x_2, y_2), the Euclidean distance between them is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



More features in your data

Example–For two points in 3D space (x_1, y_1, z_1) and (x_2, y_2, z_2), the distance is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

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DISTANCE IN KNN (OTHER DISTANCE METRICS)

Euclidean Distance– It is the most common distance metric and is used when data is dense or continuous. It's the default metric for many algorithms.

- Formula: $d(p,q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \ldots + (p_n - q_n)^2}$
- Python Snippet:

```
from scipy.spatial import distance
dst = distance.euclidean(p, q)
```